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Mapping Constellations for Transformative Futures'

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Conversation:
Mapping constellations for transformative futures

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Abstract: How might we use systemic constellation mapping to represent complex societal and multi-actor issues that describe social, climate and economic in/justices, to better understand how design can make a difference? This conversation addresses 'transformative futures' by adopting an expanded position to place the individual positionality of the designer into a constellation in which their opinions of the ways the world could or ought to be, come into tension with other conditions and actors (human and more-than-human) through the forming of 'systemic constellations'. The conversation will use bodystorming methodologies to enquire into the representation of social, climate and economic in/justices, and asks us to reflect on the implications of design as it attempts to intervene.

Keywords: Constellation; Mapping; Regenerative; Design

1. Background

Following the highly participative DRS2022 Conversation entitled 'Designing Transformative Futures' (Light et al 2022), this second iteration extended the use of bodystorming methodologies to enquire into the representation of social, climate and economic in/justices, and asked participants to reflect on the implications of design – as they understand and embody it – as it attempts to intervene.

Sarkar's portrayal of ecosystems of wicked problems (Fig 1.), presents a view of entangled economic, geo-political and ecological issues that create conditions for 'destruction of the common good' (Sarkar et al, 2023). Whilst Sarkar's model is depicted from an extremely high level and like any representation is situated and partial knowledge, many designers will

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identify with the conditions of intractable circumstances (Rittel and Webber's 'wicked problems') as we learn, research, and engage within increasingly complex circumstances. In recent years, decolonial design has developed theoretical and methodological frameworks for design to support envisioning and enacting fairer futures within complexity (Kambunga et al, 2023, Leitão & Noel, 2022). And whilst speculative design and futures thinking has enabled design to "create spaces for discussion and debate about alternative ways of being" (Dunne & Raby, 2013), there remains complexity in representing current converging issues, and challenge in rehearsing the possible pathways through which design makes constructive contribution.

At DRS2022 in Bilbao, a conversation was convened to explore the positionality of individual design researchers within a series of shared acts, paying attention to the self-fulfilling form of prediction that occurs when designers attempt to design alternative ways of being (Lockton & Ranner, 2017). Made visible through a body storming exercise asking participants to respond to questions upon 'future-making', groups of ~25 people charted their individual position within a quadrant mapped on the floor, or virtually on a Miro board for those attending online. This movement allowed participants to make visible those predictions. Responding to statements given by the facilitators, across three waves of quadrants, individuals explored the contradictions and conflicts that inhibited or enabled them to move toward future making (Fig 2.) (Light et al, 2022).

The 2022 exercise focused upon the individual design researcher's 'crisis' as they 'engage through the lens of their character and identity in complex design judgments that are always already "ethically-valenced" (Light et al, 2022). This second iteration of the conversation addressed 'transformative futures' by adopted an expanded position to place the individual positionality of the designer into a constellation in which their opinions of the ways the world *could* or *ought to* be (Gray & Chivukula, 2019) come into tension with other conditions and actors (human and more-than-human) through the forming of 'systemic constellations'. Developing such a sociodrama (Moreno 1987) allows participants to model progressing through time, exploring the impact of different conditions and decisions.

Given the intellectual and cultural experiences that attendees from the DRS community bring, the introduction of a 'distributed positionality' offered an opportunity to map complex systemic constellations, while performing collaborative designed interventions that anticipate consequences within an ecosystem (constructive or destructive), rather as a technical model might, but with the liveness that unexpected contributions from diverse perspectives, experiences and situated ethics can bring.

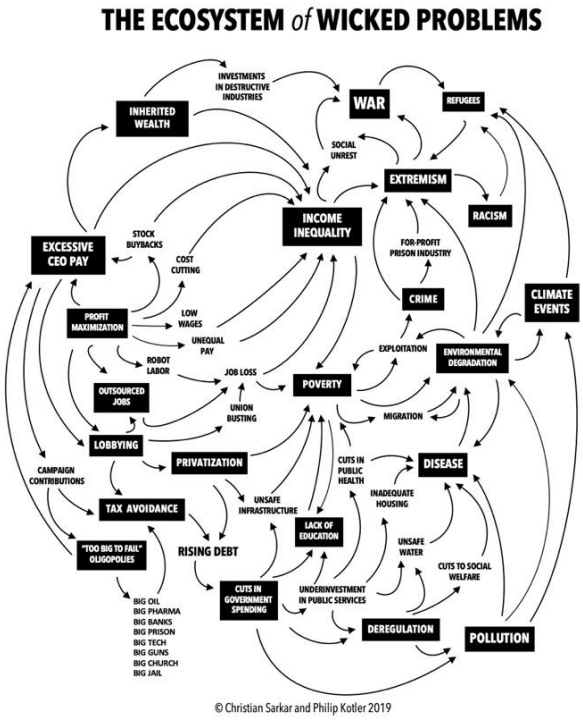


Figure 1 Sarkar's representation of the ecosystem of wicked problems that leads to the destruction of the common good. Image: Christian Sarkar & Philip Kotler.



Figure 2 Snapshot of participants at 2022 Conversation changing their position across quadrant on floor. Image: Authors.

2. Activity and group reflections

2.1 The activity

The 90 minute conversation was organised into four distinct parts: a welcome including Acknowledgement of Country, introduction, the activity itself, and then followed by short reflections from each facilitator.

The authors welcomed participants into the shared space and provided a brief introduction to the agenda. The introduction set the activity in context with the DRS2022 Conversation and explained the shift from quadrant to constellation as organising structures. With over 50 participants 'on site', and 11 online, the authors organised groups of approximately 10-12 people before finding spaces to spread out – one group online in a Miro board led by Dan Lockton, two groups moved outside the building led by Laura Forlano and Sarah Teasley, whilst two stayed inside, Chris Speed and Joe Lindley.

After the groups were formed, each facilitator guided the process. Groups were given an image of Sarkar's ecosystem of wicked problems and cards with each *element* of the ecosystem written on them, for example 'war', 'poverty', 'environmental degradation' etc.

Participants were asked to select a card and consider how they might adopt a position alongside fellow participants in the group.

Having taken a physical position within each shared space, the facilitator encouraged participants to develop a discussion upon the positions that 'make sense' of both their adopted *element*, and its relationship with others within the system. This may have been initiated by a participant that finds themselves collocated to another *element* that compromises an aspect of social, climate and/or economic justice, and sees opportunity to align with the disposition of another element. Each group was provided a single continuous paper streamer to connect elements / participants.



Figure 3 Participants in each group were asked to adopt an element from Sarkar's ecosystem of wicked problems, and co-locate themselves in relation to other elements, using a paper streamer to suggest connections. Image: Colin Gray.

Following the initial positioning of participants within each space, the request was given for one individuals toannounce a design proposition to mitigate against their 'wicked circumstance', and as a result everyone else in the group is asked to readjust their position and co-location as they interpret the implications. Conversation was encouraged to process the implications of each move. The groups were then encouraged to repeat the activity a series of times to develop a confidence anticipating the implications of actions across a complex system.

The activity concluded with a reflection on how a co-designed intervention might either ameliorate or exacerbate justice or injustice within the constellation. The activity was mirrored using Miro to include online participants.

The conversation concluded with authors / facilitators who "played back" how each constellation responded to the process, describing the how the ecosystem was represented

and enacted, and reflecting on the intra-actions across each ecosystem. Furthermore, reflections were offered on the extent to which the Systemic Constellation Mapping allowed the groups to anticipate design practices for regenerative futures that balanced social, climate, and economic justice. A series of dominate themes emerged from the feedback session.

2.2 Complexity and interconnectivity

Across reflections provided by all five facilitators, there was a consensus on the inherent complexity of the topics that Sarkar's representation includes. Participants struggled with the notion that issues such as environmental degradation, corporate pay, and climate events are not isolated but deeply interconnected. This complexity made it challenging for the groups to move beyond surface-level discussions and into actionable insights.

Laura and Sarah particularly emphasised the challenge of untangling these interconnected issues. Laura's group found it difficult to reconcile the affirmative and negative framings of challenges, "So things like environmental degradation was kind of confusing… so we weren't just the environment, we were environmental degradations. And then it's like, how does that relate to lack of education or right [to education]?". While Sarah's group struggled with the gravitational pull of complexity, which kept them tightly bunched together, "as design researchers, who have been sitting with [Haraway's] trouble quote for some time… our inclination is to see the complexity and to see the interconnectivity". This suggests a recognition that the problems they face cannot be easily categorized or solved in isolation, reinforcing the need for holistic, systems-level thinking.



Figure 4 Laura's group quickly entered discussions on how to isolate the deeply interconnected themes such as environmental degradation, corporate pay, and climate events. Image: Colin Gray.

2.3 Redefining concepts

A recurring theme was the necessity to rethink and redefine established concepts like value, capitalism, and war. Dan's group, for instance, discussed how redefining value systems—moving away from money-centric metrics—could potentially reshape broader societal structures, such as government spending and environmental policies, "if we redefine capitalism to focus on something other than money or other than measuring, and other than these types of approaches, what would it do to other stuff? So, what would it do to, you know, things around taxes and government spending, but also the effect of like, a secure income for people? Would that change things around lobbying, and probably change the way we think about profit maximisation, which would then have effects on the environment, and perhaps, you know, offering more environmental incentives would, you know, would have those, those benefits?"

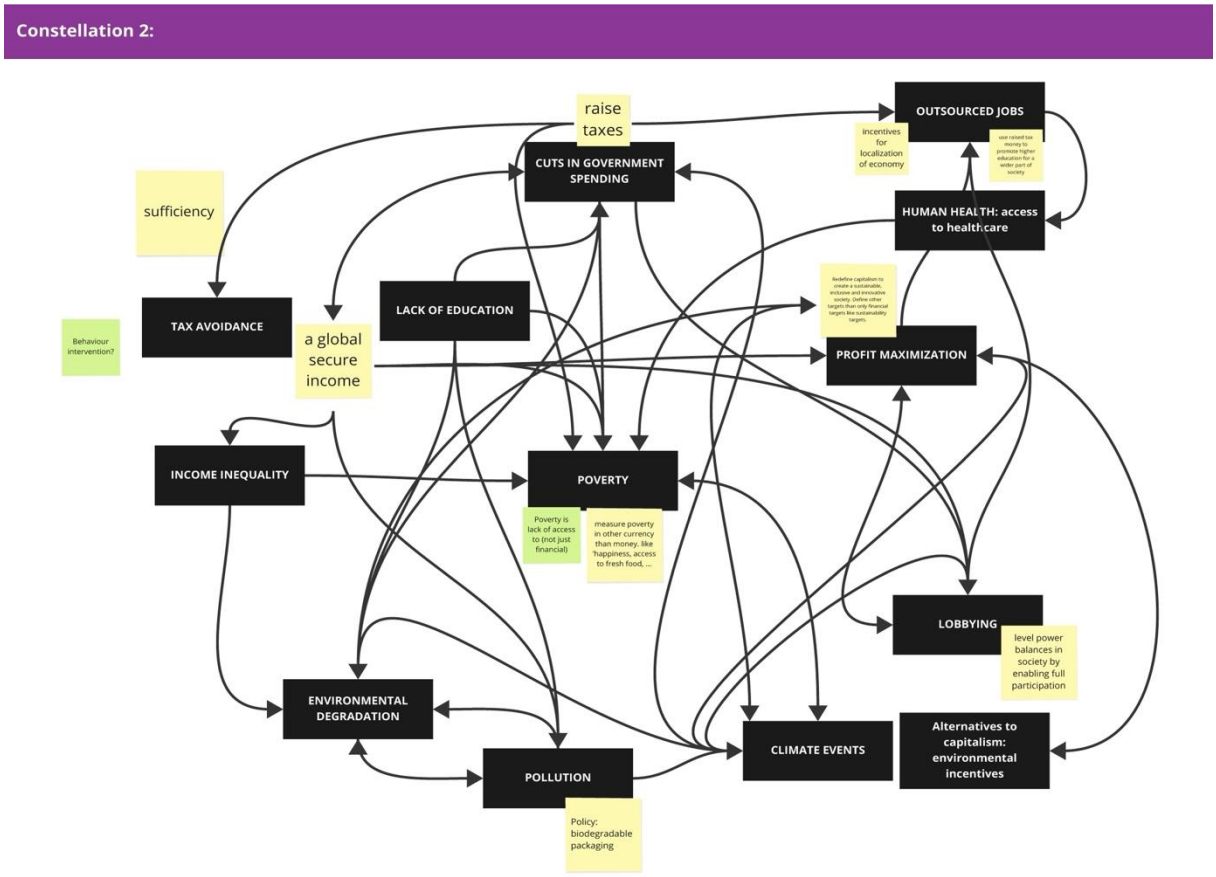


Figure 5 Dan's group used the Miro board to explore value creation and destruction when money is isolated as the primary representation of value within an ecosystem. Image: Colin Gray.

This was echoed in Chris's group's discussion, which explored the idea of recasting historical problems to address future challenges, "we talked about recasting the idea of histories, that many of the things that we present here are really historical problems, and maybe there's an opportunity to rethink and recast those futures by really going back and looking at who was misrepresented, leaning into feminist studies to think, what are feminine histories, feminist histories?".

Suggesting that the activity gave space to redefine concepts that stem from traditional paradigms that may no longer serve in addressing modern-day wicked problems. This redefinition process is seen as crucial to creating new frameworks that better align with the complex realities faced by the participants, particularly in fields such as design research.

2.4 Challenges in group dynamics

Several participants noted the difficulties in navigating group dynamics, which often stemmed from differing perspectives and the complexity of the issues at hand. Joe's group found themselves "stuck" in their initial positions, unable to move forward due to the weight of their conceptual identities and the challenge of relating these to others in the group, "It was difficult, and we were, at times, trying to know how to shape a constellation of

ourselves, but found our initial positions quite sticky. I think that there's a contrast between some sort of quite high-level conceptual identities of design researcher, like a more than human person, through to more applied problem focused things. So, finding how different nodes in the constellation related to each other across those dimensions was challenging".

This stagnation could be seen as a metaphor for the broader difficulty of design as it acknowledges the disciplines own complicity in contributing to the making of 'an ecosystem of wicked problems'.



Figure 6 Joe's group never left a tight circle which bound them to discussion. Ultimately they chose to write on to the streamer to allow them to connect themes between each other's position within an ecosystem. Image: Colin Gray.

Sarah's group highlighted the challenges of social interaction within the group, particularly around expressing disagreement. The group's respectfulness and desire to maintain relational harmony was tangible, as was the openness to explore divergences using humour. Ultimately, this tension and the multiple positions expressed revealed interesting insight into design sensibilities: "we had a fantastic discussion about the assumptions and expectations of other people in the group, that came through in the second exercise around the assumptions about people sharing views on these topics… like in this moment of sociality, as we're trying to be respectful and relational, can we say if we disagree, and how would feel unsafe if we disagreed?".

This suggests that while collaborative efforts are essential, they also require a delicate balance between respect and the willingness to challenge each other's assumptions.

2.5 Reflection on roles and responsibilities

There is a clear awareness among the participants of the evolving roles and responsibilities of design researchers in the face of societal changes. Chris's group, for example, reflected on

the tension between pursuing large academic projects and the need for more granular, iterative approaches to problem-solving, ‘there's a generation of us who are asked to pursue large projects in academia. And there was a message coming through that another generation, if it used that term, we're thinking about small movements, responsible movements, which was a really lovely moment towards knowing the scale, rather than grand challenges”. This reflects a broader debate within the design community about the scale and scope of their work—whether they should focus on grand challenges or more localised, sensitive interventions.



Figure 7 Early career researchers in Chris’ group introduced the principle that their position within design schools is less about the large project, but instead attending to the small. Image: Colin Gray.

Dan's group also touched on the potential for new roles and mechanisms in design and education, suggesting that as society evolves, so too must the disciplines that shape it, “we need new mechanisms around actually taking this sort of systemic perspective, and that might mean different roles for designers, different roles for educators… how do we design those systems?”.

This indicates a recognition that traditional roles may need to be redefined or expanded to effectively address the complex challenges of the future.

2.6 Cultural and contextual sensitivity

The discussions revealed an acute awareness of the importance of cultural and contextual sensitivity, particularly in diverse groups. Sarah’s group grappled with the implications of assumptions and expectations within a multicultural context, questioning the very idea of a unified ‘we’, “we've been talking about we, and this came out with one or two of the cards

and one or two of the responses. So, I think we had a really important critique, which points to the situatedness of any amount of complexity. This points to the challenge of finding common ground in a world where perspectives and experiences are increasingly diverse."

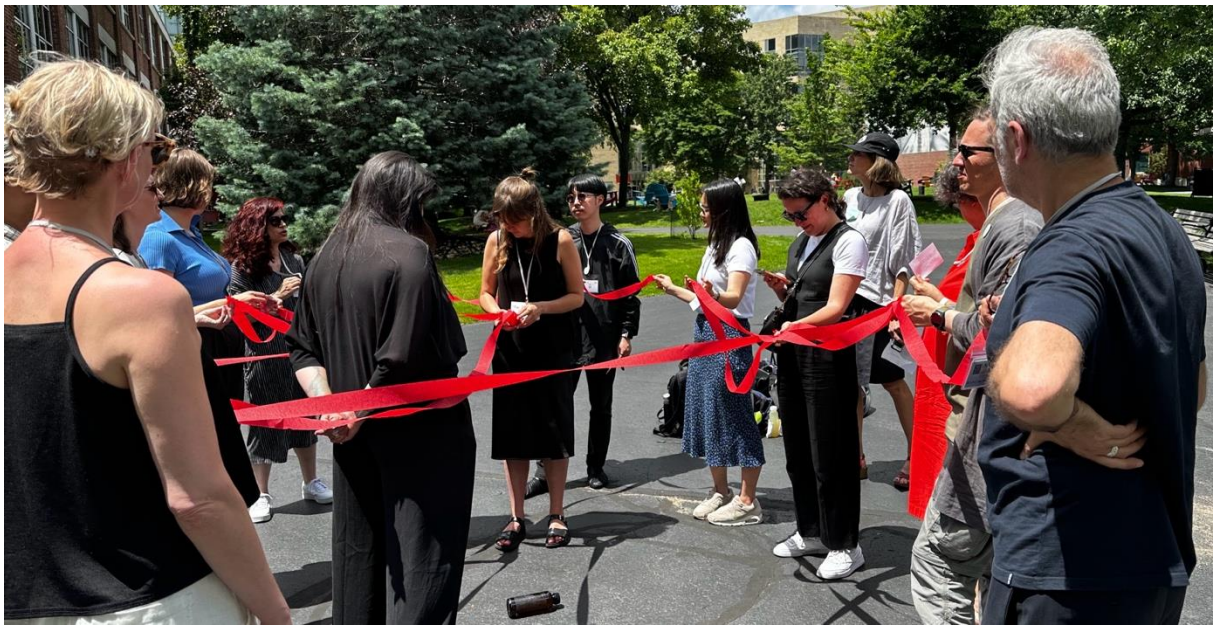


Figure 8 Sarah's group identified the trap of using the term 'we' to attend to a particular issue within Sarkar's representation of a wicked ecosystem. The conversation opened up an opportunity to discuss pluraversality over complexity. Image: Colin Gray.

This sensitivity is crucial in ensuring that the actions of design research are carefully considered to ensure equity and inclusivity. It also underscores the importance of considering local contexts and histories when addressing global challenges, as Chris's group suggested in their discussion of recasting historical problems, "the idea that war has been something that has prompted massive development of technology, cooperation, collaboration, and so could you recontextualize war as being against climate change, against, you know, the war on poverty".

3. Reflections and summary

This, the second in the series of Conversations on "Designing Transformative Futures" underscored the critical role of design in addressing the complex, interconnected challenges of social, climate, and economic injustices. By employing bodystorming methodologies, participants were able to use Sarkar's representation wicked problems, which highlighted the entangled nature of these issues and the daunting task of the design researcher intervening. The 2024 conversation reaffirmed the importance of systemic thinking in design, where traditional approaches often fall short in the face of such multifaceted challenges.

The discussions and activities pointed to the growing relevance of decolonial design and speculative futures as essential frameworks for envisioning alternative pathways in

increasingly complex environments. However, the challenges in representing and navigating these converging issues remain significant, particularly when attempting to reconcile the ethical dimensions of design work. The systemic constellation approach that provided the basis for the conversation, provided both a challenge and insights into how different conditions and decisions can influence potential futures, emphasising the need for sensitive, integrated strategies in design research.

The reflections on group dynamics and the necessity of redefining established concepts such as value co-creation and co-destruction (Speed, 2023) highlighted the evolving responsibilities of design researchers. The difficulty in moving beyond surface-level discussions to actionable insights underscored the need for more nuanced, context-sensitive approaches that can address both local and global challenges. The importance of cultural and contextual sensitivity was particularly evident (Akama et al 2024), reaffirming that design interventions must prioritise inclusivity and equity to be truly meaningful.

The conversation and its outcomes reinforce the idea that addressing the wicked problems of our time requires a rethinking of traditional design paradigms and a commitment to developing new frameworks that can better align with the complexities of a pluriversal approach to acknowledging a 'world where many worlds fit' (Cadena & Blaser 2018). The experiences of participants across both the 2022 and 2024 conversations suggest that design researchers are increasingly asked to hold on to different 'frames' simultaneously 1. the system and their place within it through a reflexive lens (future-making), and 2. their impact / affect on a system through a diffractive lens (constellation mapping). Not easy because the problem-solving positivism that design often inherits, belies both ways of doing.

Given the significant interest in both events, a future conversation might extend the series by involving 'more-than-human' participants within any structural 'device' that is used to frame the body storming. Whilst 'more-than-humans' were implicit within both conversations, participants raised the question (and opportunity) to more directly involve actors, materials and living beings that outside of the frames to date. A further opportunity might arise to run a conversation outside the academy. Although both venues at Universities in Bilbao and Boston allowed participants the privilege of isolating practice through a representational lens (quadrant and constellation), both events took place within campuses. Asking the question, what might a design research conversation that addressed the theme of Transformative Futures be like if it were to leave the academy?

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